REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

Four-slot carrier: protected external 5 V / 4 A, isolated GPIO

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM. FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

U1
Pico 2 with headers, SC1632

P_S1_IRQ	1	GP0	GP15	21	P_EXP_RESET_N
P_S1_BUSY	2	GP1	GP14	22	P_IO_OE_N
GND	3	GND	GND	23	GND
P_S2_IRQ	4	GP2	GP13	24	
P_S2_BUSY	5	GP3	GP12	25	
P_SPI_MISO	6	GP4	GP11	26	
P_S1_CS_N	7	GP5	GP10	27	
GND	8	GND	GND	28	GND
P_SPI_SCK	9	GP6	GP9	29	
P_SPI_MOSI	10	GP7	RUN	30	
P_S3_IRQ	11	GP8	GP8	31	
P_S3_BUSY	12	GP9	GP7	32	
GND	13	GND	AGND	33	GND
P_S4_IRQ	14	GP10	GP6	34	
P_S4_BUSY	15	GP11	ADC_VREF	35	
P_S2_CS_N	16	GP12	3V3_OUT	36	PICO_3V3
P_S3_CS_N	17	GP13	3V3_EN	37	
GND	18	GND	GND	38	GND
P_S4_CS_N	19	GP14	VSYS	39	PICO_VSYS
P_EXP_CS_N	20	GP15	VBUS	40	USB_VBUS

J1
SSQ-110-01-G-S

GND	1	1	6	SPI_MOSI
RADIO_5V	2	2	7	SPI_MISO
RADIO_5V	3	3	8	S1_CS_N
GND	4	4	9	S1_RESET_N
SPI_SCK	5	5	10	GND

J3
SSQ-110-01-G-S

GND	1	1	6	SPI_MOSI
RADIO_5V	2	2	7	SPI_MISO
RADIO_5V	3	3	8	S2_CS_N
GND	4	4	9	S2_RESET_N
SPI_SCK	5	5	10	GND

J9
5 V / 4 A INPUT

EXT_5V	1	1	2	GND

U2
TPS25947OLRPWR

UVLO	1	EN_UVLO	OUT	8	RADIO_5V
OVLO	2	OVLO	DVST	7	DVST
	3	AUXOFF	GND	6	GND
	4	FLT_N	ILIM	9	ILIM
EXT_5V	5	IN	ITIMER	10	

Q1
DMG2305UX-7

USB_VBUS	1	G	D	3	RADIO_5V
PICO_VSYS	2	S			

J2
SSQ-108-01-G-S

S1_BUSY	1	1	5	
S1_IRQ	2	2	6	
S1_TX_EN	3	3	7	IOREF_3V3
S1_RX_EN	4	4	8	GND

J4
SSQ-108-01-G-S

S2_BUSY	1	1	5	
S2_IRQ	2	2	6	
	3	3	7	IOREF_3V3
	4	4	8	GND

U3
AP2112K-3.3TRG1

RADIO_5V	1	VIN	NC	4	
GND	2	GND	VOUT	5	IOREF_3V3
RADIO_5V	3	EN			

J10
SSQ-120-01-G-S

P_S1_IRQ	1	1	11	P_S3_IRQ
P_S1_BUSY	2	2	12	P_S3_BUSY
GND	3	3	13	GND
P_S2_IRQ	4	4	14	P_S4_IRQ
P_S2_BUSY	5	5	15	P_S4_BUSY
P_SPI_MISO	6	6	16	P_S2_CS_N
P_S1_CS_N	7	7	17	P_S3_CS_N
GND	8	8	18	GND
P_SPI_SCK	9	9	19	P_S4_CS_N
P_SPI_MOSI	10	10	20	P_EXP_CS_N

J5
SSQ-110-01-G-S

GND	1	1	6	SPI_MOSI
RADIO_5V	2	2	7	SPI_MISO
RADIO_5V	3	3	8	S3_CS_N
GND	4	4	9	S3_RESET_N
SPI_SCK	5	5	10	GND

J7
SSQ-110-01-G-S

GND	1	1	6	SPI_MOSI
RADIO_5V	2	2	7	SPI_MISO
RADIO_5V	3	3	8	S4_CS_N
GND	4	4	9	S4_RESET_N
SPI_SCK	5	5	10	GND

J11
SSQ-120-01-G-S

P_EXP_RESET_N	1	1	11	
P_IO_OE_N	2	2	12	
GND	3	3	13	GND
	4	4	14	
	5	5	15	
	6	6	16	PICO_3V3
GND	7	7	17	
	8	8	18	GND
	9	9	19	PICO_VSYS
	10	10	20	USB_VBUS

J6
SSQ-108-01-G-S

S3_BUSY	1	1	5	
S3_IRQ	2	2	6	
S3_TX_EN	3	3	7	IOREF_3V3
S3_RX_EN	4	4	8	GND

J8
SSQ-108-01-G-S

S4_BUSY	1	1	5	
S4_IRQ	2	2	6	
S4_TX_EN	3	3	7	IOREF_3V3
S4_RX_EN	4	4	8	GND

U4
MCP23S17-E/SO

S1_RESET_N	1	GPB0	A2	15	GND
S2_RESET_N	2	GPB1	A1	16	GND
S3_RESET_N	3	GPB2	A0	17	GND
S4_RESET_N	4	GPB3	RESET_N	18	EXP_RESET_N
	5	GPB4	INTB	19	
	6	GPB5	INTA	20	
	7	GPB6	GPA0	21	S1_TX_EN
	8	GPB7	GPA1	22	S1_RX_EN
IOREF_3V3	9	VDD	GPA2	23	S3_TX_EN
GND	10	VSS	GPA3	24	S3_RX_EN
EXP_CS_N	11	CS_N	GPA4	25	S4_TX_EN
SPI_SCK	12	SCK	GPA5	26	S4_RX_EN
SPI_MOSI	13	SI	GPA6	27	
EXP_MISO_SRC	14	SO	GPA7	28	

R22
22

EXP_MISO_SRC	1	1	2	SPI_MISO

#FLG01
POWER DECLARATION: EXT_5V

EXT_5V	1	POWER

#FLG02
POWER DECLARATION: GND

GND	1	POWER

#FLG03
POWER DECLARATION: PICO_VSYS

PICO_VSYS	1	POWER

U5
SN74LVC8T245PWR

PICO_3V3	1	VCCA	GND	13	GND
PICO_3V3	2	DIR	B8	14	EXP_RESET_N
P_SPI_SCK	3	A1	B7	15	EXP_CS_N
P_SPI_MOSI	4	A2	B6	16	S4_CS_N
P_S1_CS_N	5	A3	B5	17	S3_CS_N
P_S2_CS_N	6	A4	B4	18	S2_CS_N
P_S3_CS_N	7	A5	B3	19	S1_CS_N
P_S4_CS_N	8	A6	B2	20	SPI_MOSI_SRC
P_EXP_CS_N	9	A7	B1	21	SPI_SCK_SRC
P_EXP_RESET_N	10	A8	OE_N	22	P_IO_OE_N
GND	11	GND	VCCB	23	IOREF_3V3
GND	12	GND	VCCB	24	IOREF_3V3

U6
SN74LVC8T245PWR

PICO_3V3	1	VCCA	GND	13	GND
GND	2	DIR	B8	14	S4_IRQ
P_S1_BUSY	3	A1	B7	15	S4_BUSY
P_S1_IRQ	4	A2	B6	16	S3_IRQ
P_S2_BUSY	5	A3	B5	17	S3_BUSY
P_S2_IRQ	6	A4	B4	18	S2_IRQ
P_S3_BUSY	7	A5	B3	19	S2_BUSY
P_S3_IRQ	8	A6	B2	20	S1_IRQ
P_S4_BUSY	9	A7	B1	21	S1_BUSY
P_S4_IRQ	10	A8	OE_N	22	P_IO_OE_N
GND	11	GND	VCCB	23	IOREF_3V3
GND	12	GND	VCCB	24	IOREF_3V3

U7
SN74LVC1T45DBVR

PICO_3V3	1	VCCA	B	6	SPI_MISO
GND	2	GND	DIR	5	GND
P_SPI_MISO	3	A	VCCB	4	IOREF_3V3

MP1
M3 captive-post clamp nut

MP2
M3 captive-post clamp nut

MP3
M3 captive-post clamp nut

MP4
M3 captive-post clamp nut

MP5
External cable mating plug

R20
22

SPI_SCK_SRC	1	1	2	SPI_SCK

R21
22

SPI_MOSI_SRC	1	1	2	SPI_MOSI

R20/R21: immediately after U5 SCK/MOSI outputs. R22: at U4 SO before shared MISO.
22R initial; 0R population option. Verify by oscilloscope/IBIS. No long stubs.
Each radio card has its own MISO source resistor before slot A7.

R1
267k

EXT_5V	1	1	2	UVLO

R2
100k

UVLO	1	1	2	GND

R3
340k 0.1%

EXT_5V	1	1	2	OVLO

R4
100k 0.1%

OVLO	1	1	2	GND

R5
715

ILIM	1	1	2	GND

C1
10nF

DVST_C	1	1	2	GND

C2
1uF

EXT_5V	1	1	2	GND

C3
220uF

RADIO_5V	1	+	-	2	GND

C4
22uF

RADIO_5V	1	1	2	GND

C5
22uF

IOREF_3V3	1	1	2	GND

C6
100nF

IOREF_3V3	1	1	2	GND

C7
100nF

PICO_3V3	1	1	2	GND

C8
100nF

IOREF_3V3	1	1	2	GND

C9
100nF

PICO_3V3	1	1	2	GND

C10
100nF

IOREF_3V3	1	1	2	GND

C11
100nF

PICO_3V3	1	1	2	GND

C12
100nF

IOREF_3V3	1	1	2	GND

R6
10k

PICO_3V3	1	1	2	P_IO_OE_N

R7
100k

USB_VBUS	1	1	2	GND

R8
10k

EXP_RESET_N	1	1	2	GND

R9
10k

IOREF_3V3	1	1	2	EXP_CS_N

R10
100k

SPI_MISO	1	1	2	GND

R11
1k

RADIO_5V	1	1	2	GND

R12
1k

IOREF_3V3	1	1	2	GND

R13
10k

P_EXP_RESET_N	1	1	2	GND

R14
10k

PICO_3V3	1	1	2	P_S1_CS_N

R15
10k

PICO_3V3	1	1	2	P_S2_CS_N

R16
10k

PICO_3V3	1	1	2	P_S3_CS_N

R17
10k

PICO_3V3	1	1	2	P_S4_CS_N

R18
10k

PICO_3V3	1	1	2	P_EXP_CS_N

R19
100

DVST	1	1	2	DVST_C

18/26 GPIO used; eight spare including GP26–28, all physically NC.
BUSY/IRQ/CS are direct GPIO through fixed-direction isolation, not expander GPIO.
KEY: captive M3 x 25 alignment post, see radio-slot-v0.1.md.
U4 GPA0..5 = S1/S3/S4 TX/RX pairs; GPB0..3 = S1..4 RESET_N.
No autonomous duty-cycled RX or automatic TX/RX transitions on Ebyte cards.

USB-only: Q1 off; RADIO_5V and IOREF_3V3 unpowered. External power: U2 -> radios and Q1 -> VSYS.
Q1 orientation is intentional: body diode from RADIO_5V toward VSYS. Pico onboard D1 blocks USB backfeed.
#FLG01/#FLG02 model the external supply; #FLG03 models real Q1/onboard USB-D1 ORing output.
U2 AUXOFF/FLT/ITIMER unused per datasheet. Leave ITIMER open for fastest current limiting.

H1
Captive M3 x 25 alignment post

H2
Captive M3 x 25 alignment post

H3
Captive M3 x 25 alignment post

H4
Captive M3 x 25 alignment post

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: carrier-rev-a.kicad_sch

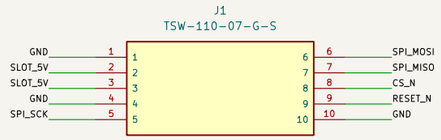
Title: Four-slot carrier: protected external 5 V / 4 A, isolated GPIO

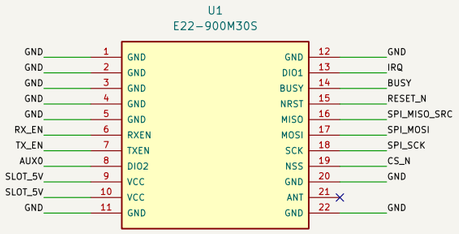
Size: A1	Date: 2026-09-17
KiCad E.D.A. 10.0.6	Rev: A-schematic
	Id: 1/1

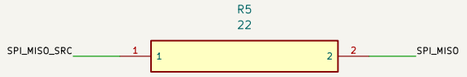
REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

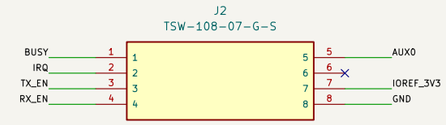
915 MHz LoRa card

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

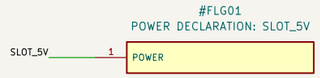


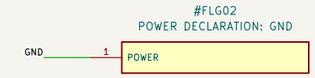


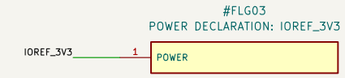


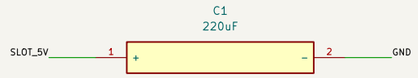


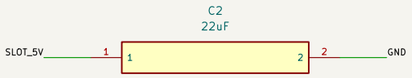
Module pad 21 ANT: NC, use module IPEX. Both pads 9 and 10 are VCC. H1 is the mandatory alignment-post hole.
DIO2 is AUX0 for a local test point only; do not connect it to a Pico GPIO.
AUX1 is reserved. DIO3 is internal TCXO control, configured to 2.2 V.

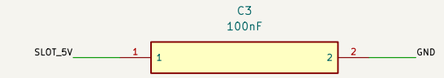


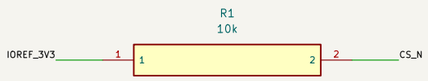


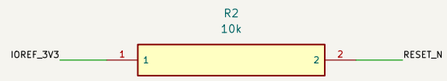


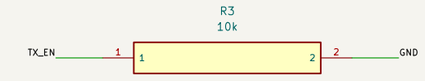


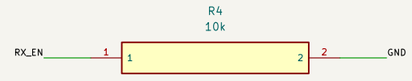














MISO: source -> 22R -> slot A7. Fit 0R option; scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
No PCB layout is authorized. Footprint and assembly evidence: see DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: e22-900m30s-rev-a.kicad_sch

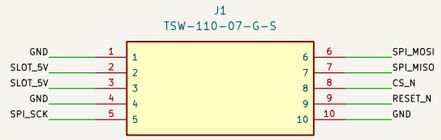
Title: 915 MHz LoRa card

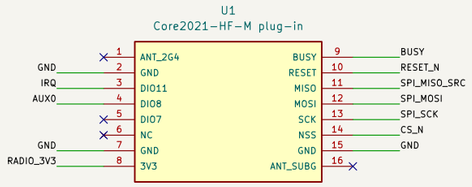
Size: A3	Date: 2026-09-17	**Rev: A-schematic**
KiCad E.D.A. 10.0.6		Id: 1/1

REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

915 MHz FLRC socket adapter

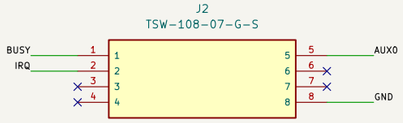
Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.



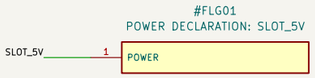


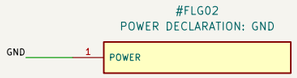


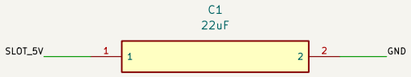


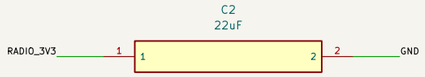


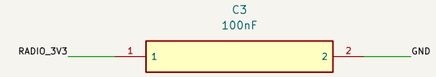
U1 represents the plug-in module, not a soldered module footprint.
Socket contacts for module pins 1, 5, 6 and 16 are isolated.
TX_EN/RX_EN slot contacts unused. No external RF switch added.
J3/J4: 15.24 mm row spacing; top/bottom numbering in FOOTPRINT_REVIEW.md.

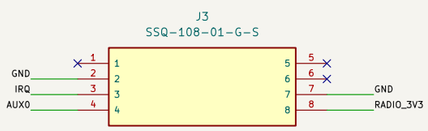


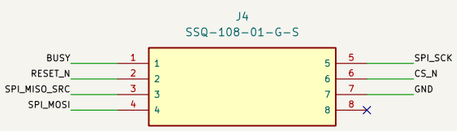


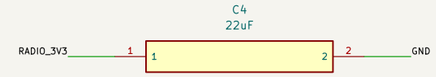


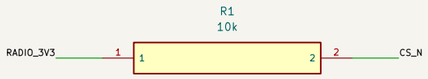


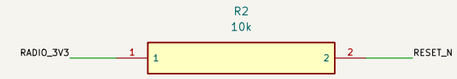


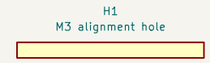












MISO: source -> 22R -> slot A7. Fit 0R option; scope/IBIS verify.
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Power declarations describe power arriving through the slot and buck output after L1.
No PCB layout is authorized. Footprint and assembly evidence: see DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: core2021-hf-m-rev-a.kicad_sch

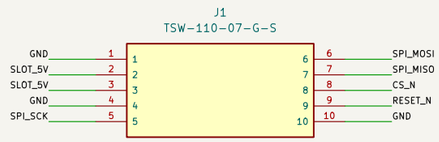
Title: 915 MHz FLRC socket adapter

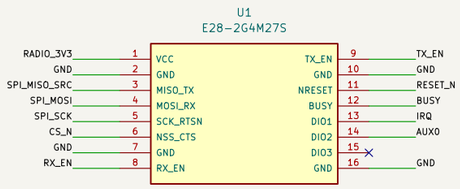
Size: A3	Date: 2026-09-17	**Rev: A-schematic**
KiCad E.D.A. 10.0.6		Id: 1/1

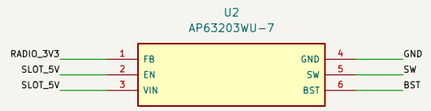
REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

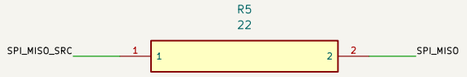
2.4 GHz LoRa or FLRC card, same hardware

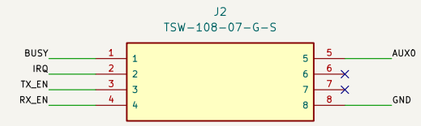
Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.



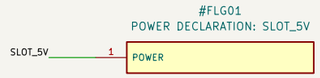




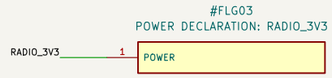


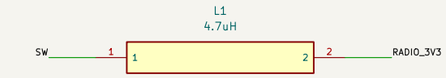


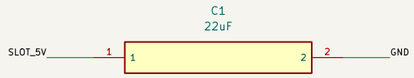
ASSEMBLY VARIANT: IPEX ONLY. Selector MUST match manual section 9.2 right image.
MCP23S17 controls TX_EN/RX_EN before RF commands. No automatic RX/TX modes.
Buck PCB layout must follow figure 25, 2 oz copper on both layers.

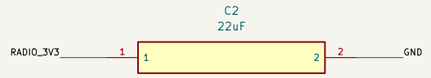


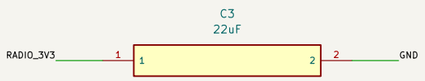


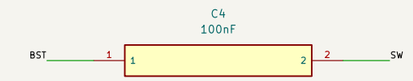


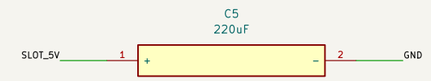


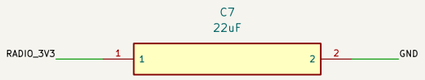


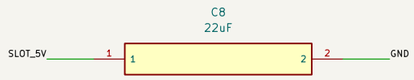


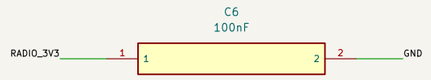


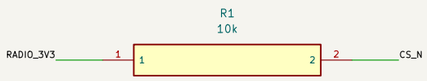


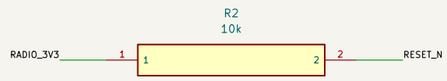


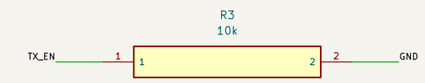


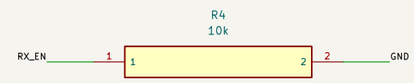


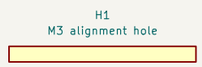












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Sheet: /
File: e28-2g4m27s-rev-a.kicad_sch

Title: 2.4 GHz LoRa or FLRC card, same hardware

Size: A3	Date: 2026-09-17	**Rev: A-schematic**
KiCad E.D.A. 10.0.6		Id: 1/1